The Contribution of the UN Decade of Ocean Science for Sustainable Development to **the Achievement of the 2030 Agenda**

The United Nations
Decade of Ocean Science
for Sustainable Development
(2021–2030)

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Foreword

The United Nations Decade of Ocean Science for Sustainable Development (the Ocean Decade) comes at a critical moment.

The growing recognition that the role of a healthy ocean must be central to sustainable development and an ocean economy was evident in the Glasgow Climate Pact outcome at COP 26 in 2021 and in the work and ambitions of the High Level Panel for a Sustainable Ocean Economy, amongst other global efforts and frameworks.

The 2022 UN Ocean Conference in Lisbon, Portugal, is the second global gathering dedicated to SDG 14 but the first featuring the Ocean Decade. The focus of the Conference is science, technology and innovation, and it is thus a crucial opportunity to further elevate the role of ocean science in not only finding solutions to climate change but also to enhance and preserve the biodiversity of the ocean, sustainably feed the population, end poverty and inequality, and stimulate a thriving and sustainable ocean economy.

To that end, this Policy Brief describes the important role of the ocean in achieving the 2030 Agenda sustainable development goals (SDG), the transversal nature of SDG 14 (life below water) and how the Ocean Decade provides the global framework to generate and use the knowledge required for sustainable ocean management and therefore a sustainable planet for future generations.

The ocean holds vast potential in providing solutions to the many challenges society faces. I invite world leaders and the global community to join forces in recognizing the importance of investments in ocean science and using this knowledge to make informed policy and management decisions.

Dr Vladimir Ryabinin
Executive Secretary
Intergovernmental Oceanographic Commission of UNESCO

Introduction

The 2030 Agenda for Sustainable Development ('2030 Agenda') defines 17 sustainable development goals (SDGs) that aim to eradicate poverty, hunger, and inequalities and set the world on a sustainable and resilient path by the end of the decade. Achieving sustainable development will require working across sectors to address multiple societal needs. Owing to the complex interactions of society, many of the SDGs are linked or serve as preconditions for achieving others. Mapping the interactions between SDGs has been explored through various analyses since the inception of the 2030 Agenda, which has proven to be an informative exercise to determine the ways to support initiatives that will have the greatest impact. This Policy Brief will focus on the transversal nature of SDG 14 (to conserve and sustainably use the oceans, seas and marine resources for sustainable development, or 'life below water') and why this SDG deserves greater attention and support from the global community.

In efforts to support the achievement of SDG 14 and the associated targets and indicators, the UN General Assembly proclaimed a Decade of Ocean Science for Sustainable Development ('Ocean Decade') to be held from 2021-2030, coinciding with the final ten years of the 2030 Agenda. The Ocean Decade was designed to provide a common framework to ensure that all countries are able to generate and support the ocean science needed to achieve the 2030 Agenda. The Intergovernmental Oceanographic Commission of UNESCO (IOC-UNESCO) was tasked to prepare an Implementation Plan for the Ocean Decade in consultation with Member States and a diversity of stakeholders. The Implementation Plan outlines ten Decade Challenges that represent the most immediate and pressing needs over the next decade (UNESCO-IOC, 2021). The resulting Decade Action Framework will guide stakeholders as they come together to co-design, co-produce and co-deliver a wide range of tangible initiatives to achieve three Decade Objectives: 1) identify required knowledge for sustainable development, 2) generate comprehensive knowledge and understanding of the ocean and 3) increase the use of ocean knowledge. Together, this framework will transform the "ocean we have" to the "ocean we want".

Recognizing that the ocean fundamentally influences many aspects of our society, achievement of SDG 14 is expected to have far-reaching positive impacts on other aspects of some of the greatest challenges we face today. Several recent reports and studies have been generated that highlight the transversal nature of SDG 14 within the 2030 Agenda, some with a focus on particular SDGs and others assessing the linkages of SDG 14 targets to other SDGs. While the Ocean Decade is championing the work to support SDG 14, it is also cross-cutting and will contribute to the achievement of many other SDGs. The purpose of this Policy Brief is to highlight how delivering transformative ocean science and sustainable development practices during the Ocean Decade will lead to a healthy and resilient ocean and greater achievement of the 2030 Agenda. Additionally, financial commitments to-date and the enabling environment required to accomplish this initiative are severely lacking and therefore, this Policy Brief brings attention to the compounding benefits of investing in ocean science. This analysis will also provide recommendations on addressing the remaining challenges and knowledge gaps to achieving the 2030 Agenda and the Ocean Decade.

The Ocean as a Driver of Human Well-being and Functioning Earth Systems

The ocean is threatened on various fronts from warming temperatures, ocean acidification, the loss of oxygen, increasing marine pollution, and an over-exploitation of finite marine resources. It is also one of the greatest assets that Earth has for supporting life and holds vast potential for solutions to many societal issues including climate change, food security, and human health (Hoegh-Guldberg et al., 2019). There are significant knowledge gaps that remain in our understanding of achieving a healthy ocean and which interventions could have the greatest benefit to society. Identifying these challenges and knowledge gaps are critical to overcoming them and unlocking solutions. Generating the necessary knowledge, enhancing the capacity to deliver this science, and increasing humanity's value of a healthy ocean around the globe is the only way to address these gaps and to make informed policy and management decisions. For the purpose of this Policy Brief, several cross-cutting themes and challenges that are relevant to a sustainable ocean, and therefore a sustainable society, will be highlighted.

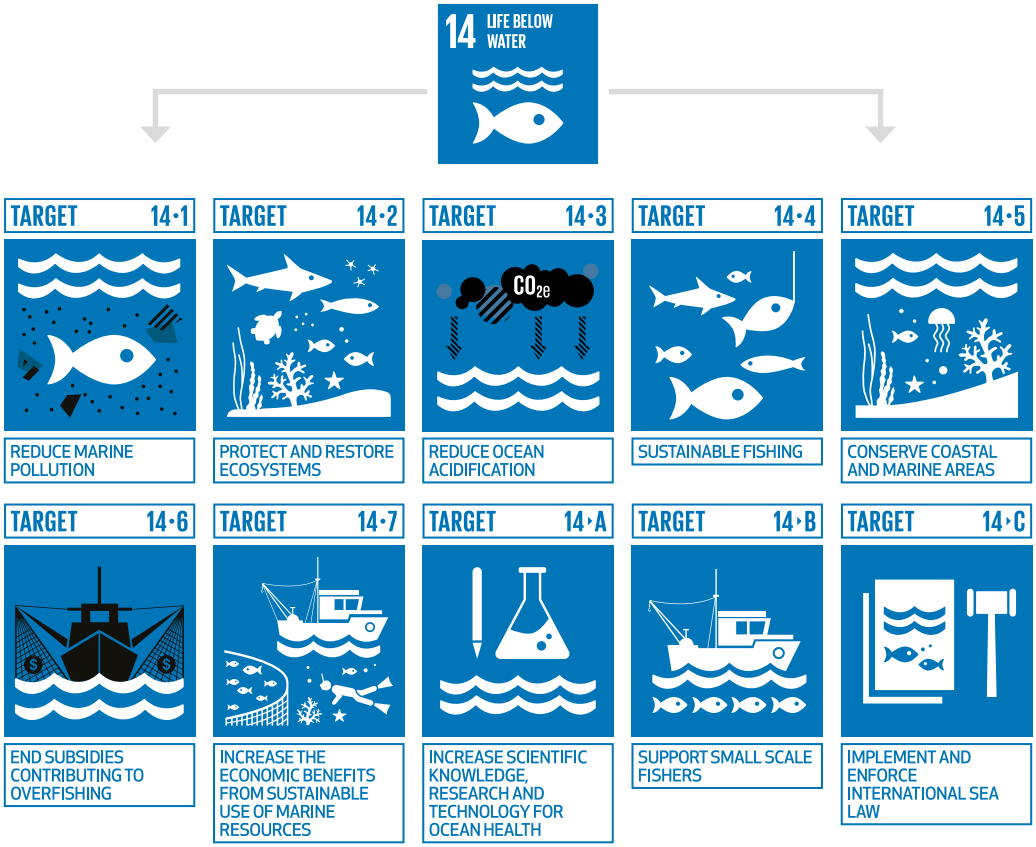
- Human-caused climate change is resulting in the deterioration of healthy marine ecosystems by absorbing excess heat and carbon dioxide from the atmosphere. The ocean also serves as the engine of Earth's climate system, with a warming ocean leading to profound impacts around the world including extreme weather events, sea level rise and marine heat waves that threaten fragile marine ecosystems (IPCC, 2019).
- The combined and cumulative impacts of ocean acidification, warming temperatures, eutrophication of coastal waters, and marine pollution are disrupting marine ecosystem structure and function, and vary in the magnitude and combination of stressors around the world (IOC-UNESCO, 2022). Therefore, individual stressors cannot be addressed in isolation.
- Over 1 billion people in developing countries rely on the ocean as their primary source of protein, with over 3 billion people globally acquiring at least 20% of their animal protein demands from the ocean (United Nations, 2017). However, unsustainable practices including overfishing, and illegal, unreported, and unregulated (IUU) fishing are decimating this important food source for the world and for future generations. These illegal fishing practices are also connected with significant human rights concerns.
- Food security is threatened for Indigenous peoples that have relied on subsistence fishing and hunting activities for centuries or longer owing to dwindling resources connected to species range shifts and marine ecosystem degradation resulting from climate change (Arctic Council, 2021).
- An estimated 680 million people are living in low-lying coastal areas, including the Small Island Developing States (SIDS), facing risks from sea level rise, storm surges, tsunamis, and other coastal hazards (IPCC, 2019). Coastlines are also becoming increasingly urban, with grey infrastructure encroaching on marine habitats which previously provided communities with some protection from the coastal threats listed (UNFCCC, 2020).
- The ocean economy is estimated to be valued between $3-6 trillion per year and is the source of over 350 million jobs around the world. Additionally, the ecosystem services the ocean provides are estimated to be over $30 billion annually (UNCTAD, 2016). Despite these valuations, the ocean has still not been fully realized, as indicated through lack of investments and inclusion in global policy frameworks, as a solution to many of the sustainable development challenges we face.
- Marine pollution, in the form of particulate matter such as microplastics or sewage and dissolved matter such as synthetic fertilisers and hormones, impacts food security, nutrition, and clean water. In areas of high agricultural runoff, increased growth of phytoplankton in stratified waters can create large oxygen depleted areas ('dead zones') within the ocean. There are increasing reports of fish caught with significant levels of microplastics, potentially reducing the nutritional benefits. Raw sewage and larger pollutants such as plastic debris also have direct impacts on those living by and visiting the coast. They pose a health threat to anyone in the ocean for leisure or livelihood. They can also have a detrimental impact on well-being which in turn may reduce marine tourism economies (World Health Organization, 2019).

Optimizing the Role of SDG 14 in the 2030 Agenda

The significance of a sustainable ocean is recognized in the context of the 2030 Agenda with a number of recent analyses highlighting the transversal role of SDG 14 as a strong influence on other SDGs (Singh et al., 2018, Sturesson et al., 2018, Winther et al., 2020, Claudet et al., 2020). These studies indicate that achieving SDG 14 only rarely had trade-offs, which were generally ones that could be adequately avoided with careful planning and working with local communities (Singh et al., 2018). The outcomes of these analyses highlight the consideration that achieving SDG 14 and other linked Goals will require transformative ocean science that is co-designed, co-produced, and co-delivered, with the inclusion of all knowledge users and generators along the ocean value chain. Bringing together these diverse actors in society to identify the critical knowledge gaps and define the research questions in ways that will result in the greatest impact, including innovative policies and new technologies, is how we can most effectively achieve SDG 14 and, as we highlight here, the overall Goals of the 2030 Agenda.

SDG 14 and Key Interactions

Figure 1. SDG 14 Targets and short descriptors.



Source: https://www.globalgoals.org/resouces

▶ SDG 14 and SDGs 1 and 2

- **Achieving SDG 1 and 2 are highly dependent on achieving SDG 14.**
- Nearly all the SDG 14 targets need to be addressed to achieve SDG 1 and 2 (Singh et al., 2018).
- The economic and social value of a healthy ocean through production of food and as a source of income cannot be overlooked in addressing SDG 1 and 2. By supporting sustainable fisheries, there are also strong linkages to SDG 8. SDG 12 is also connected to these Goals and the enhanced use of sustainable ocean resources.

— Foods harvested from the ocean are excellent sources of protein, fatty acids, and nutrients (World Health Organization, 2019). Increasing sustainable fisheries to feed the global population is a healthy source of food, contributing to SDG 3.

▶ SDG 14 and SDG 3

— The most direct linkage between these SDGs is demonstrated through the consumption of ocean-based food sources, which are generally rich in omega-3 fatty acids and essential nutrients, and are an important source of protein for the world (World Health Organization, 2019). There is also a growing industry in marine-derived pharmaceuticals and genetic resources that are being used for medicinal and other health-related purposes (Jaspars et al., 2016).

— Research has also shown that proximity of communities to healthy marine and coastal spaces has positive impacts on mental health and general well-being (World Health Organization, 2019)

— There are also negative associations with human health and well-being that are linked to marine pollution. For example, the growing use of antibiotics by society has been linked to the potential emergence of antibiotic-resistant organisms in seafood and marine mammals (Rose et al. 2009, Pepi and Ocardi 2021). Additionally, the impact of microplastics in the ocean and entering the food chain is a growing area of concern for human health (Smith et al., 2018).

▶ SDG 14 and SDG 6

— SDG 6 is strongly linked to SDG 14 through activities to reduce marine pollution (target 14.1), particularly through mitigating land-based activities and upstream sources that have negative impacts on coastal and marine ecosystems. They link other downstream effects through access to energy (SDG 7) and sustainable urban and industrial development (SDG 11) (Schmidt et al., 2017).

▶ SDG 14 and SDG 13

— Climate action is not possible without ocean action. The concept of the Ocean-Climate Nexus articulates the connection between the ocean and climate, spanning from the ocean's role in regulating climate, absorbing excess heat and carbon dioxide, to the negative effects of climate change on marine ecosystems (Hoegh-Guldberg et al., 2019). New marine technology and innovation will also be critical to renewable energy (SDG 7) and carbon dioxide removal (CDR) and sequestration. The recent IPCC assessment indicates that we are not on track to limit warming to 1.5 degrees as set out in the Paris Agreement unless we curb fossil fuels immediately and explore ways to remove carbon dioxide from the atmosphere (IPCC, 2022). The ocean will be key to the success of the Paris Agreement and achieving SDG 13.

— By increasing sustainable sources of food from the ocean, we can limit the demands of land-based agriculture which contributes substantially to climate change, which also contributes to both SDG 2 and SDG 13.

▶ SDG 14 and SDG 15

— The land and sea are interconnected systems. Restoring and protecting coastal ecosystems, such as mangroves and wetlands, enhances habitat for marine life by mitigating nutrient and sediment pollution, preventing coastal erosion and the reduction of terrestrial habitats.

▶ SDG 14 and SDG 16

— Sustainable development and management of the ocean and marine resources requires participation by all stakeholders, improving decision-making, and accountability of institutions. By building and strengthening the necessary relationships across the ocean value chain to improve ocean governance, access to resources, and ending unsustainable fishing practices, progress towards SDG 16 can be achieved (Schmidt et al., 2017).

TRANSVERSAL INTERACTIONS

Sustainable fisheries will have far-reaching benefits across many SDGs	There are co-benefits to all SDGs by increasing economic benefits for SIDS and LDCs, under the appropriate social and ecological conditions.	There can also be trade-offs among SDGs, if not carefully considered
Human rights issues can be addressed by ending IUU fishing, including ending modern slavery (SDGs 8 and 16), and enabling food security and ending malnutrition (SDG 2) (Schmidt et al., 2017). Sustainable fisheries and aquaculture will create decent work and opportunities for women, particularly in SIDS and LDCs, as they account for a substantial part of the workforce when taking into account processing and marketing (FAO, 2013).	Increasing economic benefits to SIDS and least developed countries for sustainable marine uses (SDG 14.7) was determined to have positive relationships across all SDGs. Eliminating overfishing, and illegal and destructive fishing practices (SDG 14.4) has also been found to be a necessary pre-condition for achieving the largest number of other SDG targets (Singh et al., 2018, Claudet et al., 2020).	SDG 14.5 has shown negative relationships with the greatest number of SDGs, including SDG 1, 10 and 16, which were largely related to issues of inequality, access to resources, and conflicts that may arise from interactions where stakeholders are not properly engaged (Singh et al., 2018). Co-designed MPAs, informed by ocean science, will see benefits extend beyond MPA boundaries and an overall increase of sustainable marine resources (Di Franco et al., 2016).

THE SCIENCE WE NEED

The science we need includes filling critical knowledge gaps and includes:

Understanding the ocean-climate nexus: The role of the ocean has been lacking from global climate policy frameworks. The recent IPCC's Sixth Assessment Report indicated that we need to develop and use CDR and sequestration technologies to combat the worst impacts of climate change (IPCC, 2022). This will require significant investment and resources to understand the complex ocean-climate nexus and develop safe and scalable technologies that maintain the balance of marine ecosystems (National Academies of Sciences, 2022).

Observing the ocean: Models are continuously improving with enhanced and expanded global ocean observation systems, yet there are still critical coverage gaps. We need new technology and innovation that can deliver these observations reliably and quickly. This includes near-real time digital representations of the ocean ('digital twin'), especially one that is accessible and allows decision-makers to inform global and national policy frameworks based on the best available science (Mercator Ocean International, 2022).

Multiple stressors on marine ecosystems: Understanding the impact of multiple stressors on marine ecosystems is an active area of research, where the severity of impact may differ from one location to another, and the combination of stressors has varied impacts. There is a need for increased ocean literacy on this issue in addition to improving sensors, technology and standardized methods to monitor these stressors (IOC-UNESCO, 2022).

Co-designed marine protected areas: Further research is needed to advance our understanding on marine protected areas (MPAs) and how to co-design them in ways that do not cause detriment to the local communities by limiting access to critical resources (Schmidt et al., 2017). There is potential for innovative technologies and ocean science that can inform the design of dynamic MPAs that are tailored to allow access by local communities at certain times of the year (Ortuño Crespo et al., 2020). These research questions will be critical to developing the most effective policy regulation and decision-making (United Nations, 2017).

Risks and solutions for coastal communities in response to climate change: Improved modeling and observations of the coastal zone, and access to reliable data, are needed to inform mitigation and adaptation strategies. Understanding the related socio-economic impacts requires dedicated research and resources for transdisciplinary and participatory science. Comprehensive risk management requires input from advanced technologies, such as a digital twin of the coastal ocean, to develop scenarios, and minimise loss and damage (UNFCCC, 2020).

Technologies for the marine transportation sector: Reducing greenhouse gas emissions from the marine transportation sector is a critical component to addressing climate change while supporting the global economy (Hoegh-Guldberg et al., 2019). This will require increased collaboration across a multitude of diverse actors from the ocean value chain including ocean science, engineering, private industry, technology sectors, and local communities. There are critical gaps in the relationships necessary among industry experts and policy makers to ensure the changes in technology required are implemented in a realistic, but timely fashion.

Identifying sources of marine pollution and impacts on human health and well-being: Scientists and resource managers still struggle to identify, mitigate and regulate some of the most pervasive sources of marine pollution, particularly microplastics. Research on the sources and pathways of pollutants is an ongoing priority. Greater regulatory action must be taken to ensure equitable solutions are implemented (World Health Organization, 2019).

2030 AGENDA FOR SUSTAINABLE DEVELOPMENT

THE OCEAN WE WANT

The ocean we want embraces this science and includes:

Ocean-based solutions to climate change: The ocean can be harnessed for innovative solutions to climate change, particularly through its ability to sequester carbon and provide clean, renewable energy sources. Protecting and restoring seagrass meadows and mangroves are examples of blue carbon assets with great potential to contribute towards ocean-based CDR and sequestration, when accompanied by reductions in greenhouse gas emissions. Enhancing capacity to quantify, protect and restore blue carbon assets around the world and incorporate them into nationally determined contributions (NDCs) will be critical to achieve the goals of the Paris Agreement. Restoration of these ecosystems also enhances other ecosystem services, such as improved habitat for marine species (National Academies of Sciences, 2022).

Sustainable blue foods and climate resilient fisheries: Ecosystem-based fisheries management and participatory processes that include partnerships across sectors have shown to be the most effective (Karr et al., 2017). Sustainable blue foods can also provide additional economic benefits and a safer working environment. More stringent and informed policies to end IUU fishing and harmful fishing subsidies will support this (Bank et al., 2022, United Nations, 2021, World Health Organization, 2019).

Nature-based solutions for coastal resilience: The use of nature-based solutions and innovative technologies to rethink the way communities develop infrastructure will be crucial to climate mitigation and adaptation (UNFCCC, 2020).

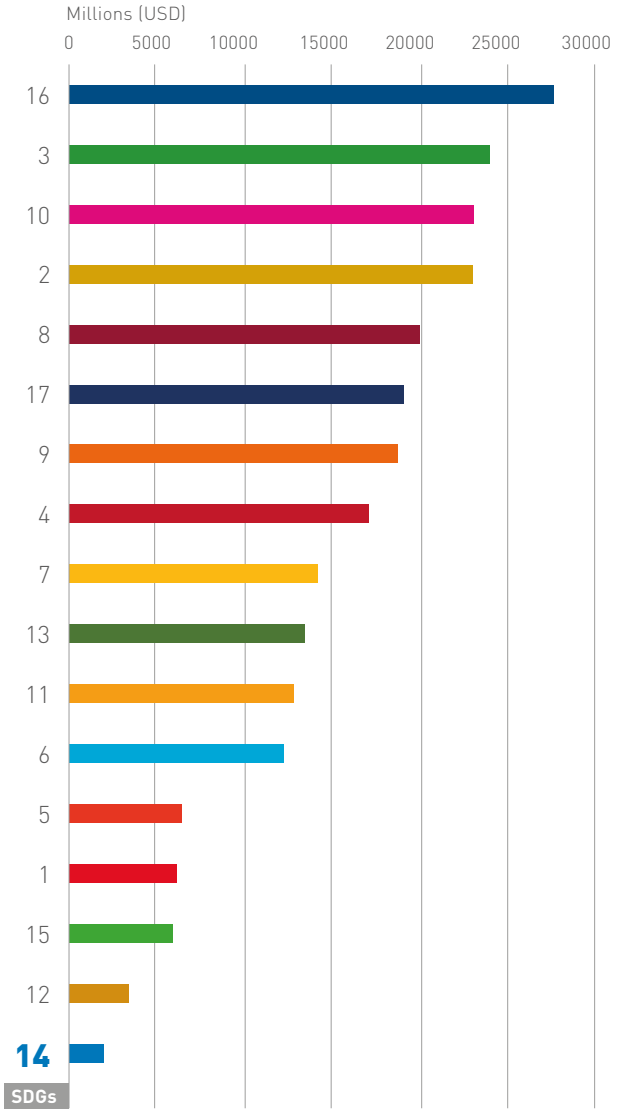
Sustainable blue economies focusing on a healthy ocean: A sustainable blue economy can make significant contributions towards sustainable development, including developing offshore renewable energy, decarbonizing maritime transport, and creating a circular economy through innovative infrastructure and gear associated with shipping, fishing and other offshore industries.

Human health and well-being: Sustainable blue foods are excellent sources of essential nutrients, contributing to overall human health and well-being, and can ease pressure on land-based agriculture (World Health Organization, 2019). Embracing Indigenous stewardship of marine resources has shown to be successful in preserving biodiversity, which also supports food security, human rights, and sovereignty of indigenous peoples around the world (FAO, 2021).

Barriers to Achieving SDG 14

While it is critical to identify the knowledge gaps and the ocean science that can contribute to addressing these challenges, there are obstacles to overcome in order to successfully deliver transformative solutions for society. The creation of an enabling environment, through developing capacity, mobilizing resources, and building support for national and global policies is fundamental to the success of the 2030 Agenda. As highlighted throughout this Policy Brief, there is tremendous and far-reaching benefit to sustainable development of the ocean. Despite this recognition, there are concerns related to funding, systemic barriers, political engagement, and methodologies that require dedicated and immediate attention.

Figure 2. Contributions from the top 25 official development assistance (ODA) providers by each SDG in 2019. Data from the SDG Financing Lab.



Source: OECD, 2022

- **SDG 14 has received the least amount of funding compared to other SDGs and ocean science makes up just a small part of this funding**
  - The lack of philanthropic funding and official development assistance (ODA) towards supporting SDG 14 relative to the other SDGs is concerning (Singh et al., 2018, Scotland, 2022, OECD, 2022).
  - Data collected by OECD through the SDG Financing Lab has determined that SDG 14 remains the least funded SDG as of 2019 from the top 25 donating Foundations (Figure 2); (OECD, 2022). This represents 0.01% of all SDG funding from development finance and 0.56% of all philanthropic finance since 2016 (Scotland, 2022).
  - These 25 Foundations provided 67% of all funding for SDG 14 and it is estimated that their contribution to marine conservation is roughly equal to that of ODA from both bilateral and multilateral aid agencies (IOC-UNESCO, 2020b).
  - Traditional funding schemes do not always align with the needs of multidisciplinary research (Sturesson et al., 2018, Claudet et al., 2020). There is a need for the development of blended funding models to support transdisciplinary, co-designed ocean science.
  - Despite the growing awareness of the strong linkages within the ocean-climate nexus, the lack of investment in ocean-based solutions to climate change are inadequate (Scotland, 2022).
  - The Global Ocean Science Report indicates that only 1.7% of national research budgets are allocated to ocean science (IOC-UNESCO, 2020a), shining a light on the lack of resources dedicated to the science that is critical to addressing the knowledge gaps of SDG 14.
  - This challenge highlights the need for increased investments and sustainable financing to support SDG 14. Foundations have shown to have the ability to take greater risks and fund emerging issues in the ocean science sector, leveraging partnerships and working across borders (IOC-UNESCO, 2020b). **There is a critical role for these sources of funding during the Ocean Decade to achieve the 2030 Agenda (Figure 3).**

- **Transdisciplinary approaches to ocean science requires a systemic change to framing problems, identifying resources, and disseminating results**
  - Generating the transformative knowledge needed for ocean action requires time for participatory research methods including proper data collection, analysis, reporting, and building relationships (Sturesson et al., 2018). Conventional academic reporting processes require time for preparing and submitting research for publication in peer-reviewed disciplinary journals. These rigorous processes are vital to the scientific method and must continue, but we also need mechanisms that can translate science into action on short

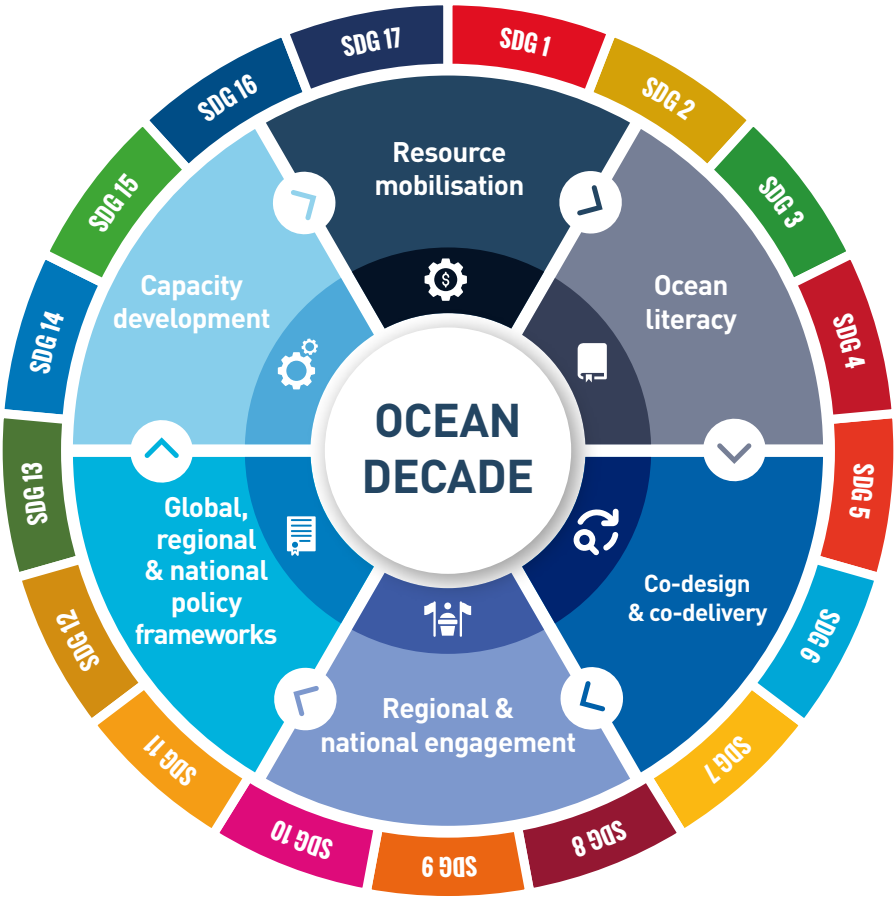
timescales and better address the increasingly complex interactions of societal issues such as climate change (Lawrence et al., 2022).

- **Research to achieve SDG 14 must go beyond the identified targets and indicators (Sturesson et al., 2018)**
  - The planning phase of the Ocean Decade brought together the global ocean science community which resulted in the ten Decade Challenges that expand beyond the targets of SDG 14 to address the most pressing needs for sustainable development of the ocean. These challenges (and therefore priorities for investments, research, etc.) are not static and will be updated throughout the Decade. Therefore, the Ocean Decade framework allows for achievement of and impact well beyond the SDG 14 targets (Figure 3).

- **There must be support and political will to identify needs, engage with the ocean science community, and to fund and implement solutions (Sturesson et al., 2018, Claudet et al., 2020)**
  - The ocean has only recently started to receive recognition in global climate policy frameworks as a solution to climate change
  - This barrier highlights the need for co-designed transformative science. The ocean science generated through the Ocean Decade encourages the engagement of policy and decision-makers (and all stakeholders of the ocean value chain) at each stage of a project so that there is enhanced uptake by society and the knowledge is used. This concept is a central tenet to the Ocean Decade (Figure 3).

- **The SDG 14 targets were adopted without a standardized monitoring approach**
  - In order to address the marine pollution target, there have been documented difficulties with standardized measurements particularly of microplastics in the ocean (Schmidt et al., 2017). There is a need for more research on sources, pathways and impacts on marine ecosystems. We also need more research on various interventions to deal with marine pollution, through policy development and coordinated action (World Health Organization, 2019).
  - Despite the inclusion of ocean acidification issues in global policy frameworks and the political lexicon, there is still a great need for capacity development when it comes to monitoring ocean acidification and literacy on this issue for society.

Figure 3. Roadmap for achieving the 2030 Agenda within the Ocean Decade.



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The Contribution of the Ocean Decade to the 2030 Agenda

Transformative Ocean Science for Addressing Challenges and Knowledge Gaps

The Ocean Decade has been designed in recognition of the interactions that exist between the ocean and global socio-ecological systems. The Ocean Decade societal outcomes recognize that a protected and functioning ocean system is essential for achieving broader societal goals. The three objectives of the Ocean Decade aim to identify critical knowledge gaps, to co-design and generate the necessary knowledge to inform global policy frameworks, and to increase the uptake of this knowledge by society which will lead to an overall improvement in society's relationship with the ocean. This behavioral change at the societal level will be critical to achieving SDG 14 and ultimately the 2030 Agenda. These strategic objectives have been developed to achieve the "ocean we want" via high-level scientific actions, referred to as 'Decade Actions'.

Ten Decade Challenges to Achieve the 2030 Agenda

After the formal proclamation of the Ocean Decade, the ocean science community responded by coming together during the planning phase to identify the most pressing challenges society faces to achieve sustainable use of the ocean. These ten challenges provide mechanisms for addressing the SDG 14 targets, but also go well beyond these to address issues from the perspective of the ocean science community and contribute to achieving the other Goals. By design, all ten of the Ocean Decade Challenges will support achievement of SDG 14. The Decade Actions endorsed within the first year are tackling each of these Challenges and will inform the needs for future calls for Decade Actions (Figure 4). Throughout the Decade, these ten challenges may be modified to reflect the rapidly changing environmental conditions. The Decade Action Framework will also contribute directly to achieve SDG 14 itself, which include reporting mechanisms for specific SDG 14 indicators.

Figure 4. Distribution of Challenges addressed by the current cohort of endorsed programmes, as reported by the programmes.

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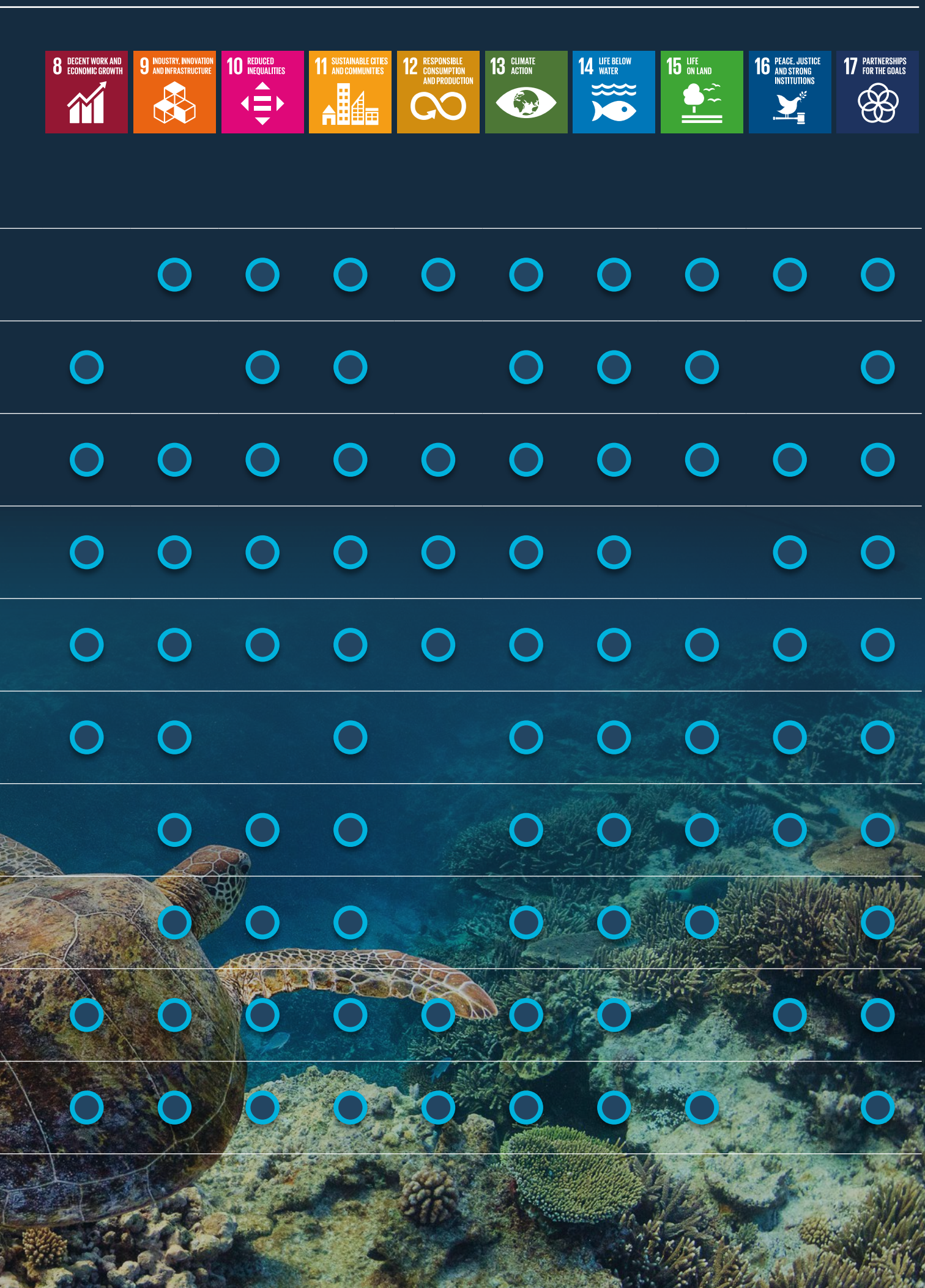
Mapping the Ocean Decade Challenges to the Sustainable Development Goals.

	1 NO POVERTY	2 ZERO HUNGER	3 GOOD HEALTH AND WELL-BEING	4 QUALITY EDUCATION	5 GENDER EQUALITY	6 CLEAN WATER AND SANITATION	7 AFFORDABLE AND CLEAN ENERGY
1 MARINE POLLUTION		●	●		●	●	
2 ECOSYSTEM RESTORATION	●	●	●			●	
3 SUSTAINABLE BLUE FOODS	●	●	●		●	●	
4 SUSTAINABLE BLUE ECONOMY	●	●	●	●	●		●
5 OCEAN-CLIMATE NEXUS	●	●	●		●	●	●
6 COMMUNITY RESILIENCE			●	●	●	●	●
7 GLOBAL OCEAN OBSERVATIONS			●	●			●
8 DIGITAL TWINS				●			
9 SKILLS, KNOWLEDGE AND TECHNOLOGY FOR ALL	●		●	●	●	●	
10 RELATIONSHIP WITH THE OCEAN	●	●	●	●	●	●	●


8 DECENT WORK AND ECONOMIC GROWTH
9 INDUSTRY, INNOVATION AND INFRASTRUCTURE
10 REDUCED INEQUALITIES
11 SUSTAINABLE CITIES AND COMMUNITIES
12 RESPONSIBLE CONSUMPTION AND PRODUCTION
13 CLIMATE ACTION
14 LIFE BELOW WATER
15 LIFE ON LAND
16 PEACE, JUSTICE AND STRONG INSTITUTIONS
17 PARTNERSHIPS FOR THE GOALS

CHALLENGE 1:
UNDERSTAND AND BEAT MARINE POLLUTION

- **SDG 14:** This challenge will specifically address target 14.1 through the identification of sources, prevention strategies and efforts to significantly reduce marine pollution, including marine debris and nutrient pollution. These efforts must be inclusive of land-based activities to be effective. The data, knowledge and capacity needed to inform indicator 14.1.1 on coastal eutrophication is being generated, for example, through activities of the endorsed Global Ocean Oxygen Decade (GOOD) Programme.
- **Other Key SDG interactions:**
  - **SDG 9 -** Identifying sources and developing intervention strategies will require a change in industrial and waste management processes. The Ocean Decade encourages transformative science working across all actors on the ocean value chain. Partnerships with private industry will be critical to meaningful changes in policy and practice.
  - **SDG 12 -** Work on Challenge 1 can contribute to target 12.4, which aims to achieve the environmentally sound management of chemicals and all wastes throughout their life cycle, in accordance with agreed international frameworks (e.g., the International Convention for the Prevention of Pollution from Ships (MARPOL) and the United Nations Convention on the Law of the Sea Treaty (UNCLOS)), and significantly reduce their release to marine and coastal environments in order to minimize their adverse impacts on human health and the environment.
  - **SDG 3 and SDG 6 -** Identifying and containing the sources of marine pollution will also contribute towards improving human health and well-being by minimizing exposure to toxic substances and by preventing upstream sources of pollution, clean water and sanitation will be reinforced.

Case Study of Decade Actions: Tackling Marine Pollution in Urban Coastal Environments

Programme: The Global Estuaries Monitoring (GEM) Programme

Decade Challenges: 1, 2, 6, 7, 9

Top 3 SDGs: 6, 11, 14

The Global Estuaries Monitoring (GEM) Programme aims to develop a global monitoring network to monitor environmental contaminants (e.g., pharmaceutical residues, emerging pollutants of concern, microplastics, pathogens) in major urbanised estuaries worldwide. GEM will facilitate capacity building for monitoring through provision of training for standard sampling and analytical methods. GEM aims to reveal the pollution situation around the globe in estuaries, identify those that require attention and improvement, recommend priority contaminants for control, and promote best practices to combat the pollution problems thereby achieving cleaner estuaries for all.

CHALLENGE 2:
PROTECT AND RESTORE ECOSYSTEMS AND BIODIVERSITY

- **SDG 14:** Focusing on the science to support the protection and restoration of marine ecosystems addresses several SDG 14 targets and indicators. This Challenge includes research efforts dedicated to understanding and mitigating the effects of multiple stressors on marine ecosystems. The Ocean Acidification Research for Sustainability (OARS) Programme will support efforts to establish standardized methodologies, reporting, and the capacity development to ensure coordinated global reporting on SDG 14.3.1 (Average marine acidity (pH) measured at agreed suite of representative sampling stations). The IOC-UNESCO, the UN agency responsible for reporting progress on indicator 14.3.1, is co-leading the OARS Programme in order to achieve this. Activities that support the design and development of marine protected areas through science-informed decision-making will also contribute to indicator 14.5.1 (Coverage of protected areas in relation

to marine areas). Monitoring the biodiversity and resilience of marine life will be addressed through initiatives of programmes such as Marine Life 2030, which will coordinate observation networks, methods, and knowledge sharing that will contribute to target 14.2 and can translate into support for countries using ecosystem-based approaches (SDG 14.2.1).

- **Other key SDG Interactions:**
  - **SDG 11 –** With UNESCO marine world heritage sites recognized as priority sites, the Ocean Decade will strengthen efforts to protect and safeguard the world's cultural and natural heritage (Target 11.4). The UNESCO Marine World Heritage programme and the Ocean Decade jointly launched an ocean science roadmap in 2021 to outline the actions needed to achieve this goal, including through technology, capacity development, and investments during the Ocean Decade (https://whc.unesco.org/en/oceanscience).
  - **SDG 12 –** Protecting and restoring marine ecosystems and biodiversity specifically addresses SDG 12.2, which promotes the sustainable use and efficient management of natural resources. Marine ecosystems are often seen by society for the services they can provide. Proper protection and management will ensure that these ecosystems thrive while still providing these valuable services.
  - **SDG 1 and 2 -** Healthy marine ecosystems are a precondition for addressing Challenge 3 (sustainable blue foods). Sustainable fisheries will be a critical component to achieving both SDGs 1 and 2.

Case Study of Decade Actions: Addressing SDG 14.3 through the OARS Programme

Programme: Ocean Acidification Research for Sustainability (OARS)

Decade Challenges 2, 5, 7, 8, 9

Top 3 SDGs: 13, 14, 17

Ocean Acidification Research for Sustainability (OARS) is providing society with observational and scientific evidence needed to sustainably identify, monitor, mitigate and adapt to ocean acidification; from local to global scales. It will foster the development of ocean acidification science including impacts on marine life and sustainability of marine ecosystems in estuarine-coastal-open ocean environments. OARS will build on the great work of the Global Ocean Acidification Observing Network (GOA-ON). The aim is to achieve SDG target 14.3 'Minimize and address the impacts of Ocean Acidification, including through enhanced scientific cooperation at all levels'. Key components include: 1) Enhancing regional collaborative efforts, 2) Coordination of capacity building in science, 3) Co-design and implement observation and research to address the threat of ocean acidification, and 4) Communication and delivery of the outputs to policy makers and communities.

Case Study of Decade Actions: Supporting SDG 14 Target 14.2

Programme: Marine Life 2030

Decade Challenges: 2, 9

Top 3 SDGs: 4, 14, 17

Marine Life 2030 will establish a globally coordinated system to deliver actionable, transdisciplinary knowledge of ocean life to those who need it, promoting human well-being, sustainable development and ocean conservation. Marine Life 2030 will unite existing and frontier technologies and partners into a global, interoperable network and community of practice advancing observation and forecasting of marine life. This network of networks will link technical, management and policy stakeholders to build and exchange capacity for advancing society's grand challenges of managing activities for a healthy and resilient ocean and the vibrant and healthy society that depends on it.

CHALLENGE 3:
SUSTAINABLY FEED THE GLOBAL POPULATION

- **SDG 14:** Through innovative approaches to ecosystem-based management and science informed policies to end illegal fishing and overfishing, this challenge will contribute towards SDG 14.2. The Ocean Decade will support the science to understand maximum sustainable fish yields based on biological characteristics, and to use this data to inform regulations on the proportion of fish stocks within biologically sustainable levels (SDG indicator 14.2.1).

- **Other key SDG Interactions:**
  - **SDG 1 –** Small-scale fisheries are an important source of income, especially in many SIDS and LDCs. Estimates suggest that more than 90% of fishers from SIDS and LDCs participate in small-scale fisheries and more than half of the world's fish catch is being produced by small-scale fisheries, with 90–95% of this catch being consumed locally (FAO, 2013). Sustainable blue foods provide economic opportunities for SIDS and LDCs, and particularly women, that will contribute to addressing Targets 1.1 and 1.2.

— **SDG 2 –** Sustainable blue foods will be essential to achieving SDG 2 and the associated targets. For example, in order to end hunger and ensure access by all people to safe, nutritious and sufficient food all year round (SDG 2.1), sustainable fisheries must be incorporated. The Ocean Decade has endorsed several Decade Actions that will provide the knowledge needed to make informed decisions about best management practices, the creation of marine protected areas, and maintaining sustainable fish stocks. Importantly, this Challenge can contribute towards the efforts to increase the productivity and incomes of small-scale, artisanal fishers, in particular women, indigenous peoples, including through secure and equal access to marine resources and inputs, knowledge, financial services, markets and opportunities for value addition (SDG 2.3).

— **SDG 3 –** Sustainable blue food provides a healthy source of nutrition that support good health and well-being. The consumption of blue foods contributes to SDG targets 3.1 (reducing global maternal mortality) and 3.2 (reducing neonatal mortality), by providing essential micronutrients, such as omega-3s, zinc, iron and vitamins A, D, and B12, that are shown to reduce infant and maternal mortality. Blue foods have also been linked to lower risks of heart disease and other non-communicable diseases (United Nations, 2021).

Case Study of UN-led Decade Actions: The Food and Agricultural Organization of the United Nations (FAO) Ecosystem Approach to Fisheries

Programme: The EAF-Nansen Programme

Decade Challenges: 1, 2, 3, 4, 5, 9, 10

Top 3 SDGs: 2, 14, 13

Since 1975, the EAF-Nansen Programme has been a collaborative platform for ocean science, fisheries management and capacity development, continuously evolving to address emerging challenges. In its current phase (2017-2023), the Programme is assisting 32 partner countries and four regional organizations in Africa and the Bay of Bengal to improve their capacity to implement the Ecosystem Approach to Fisheries (EAF), taking into consideration the impacts of climate change and marine pollution. The longterm objective is to improve food and nutrition security in partner countries. The programme comprises three key areas of work: (1) Science and Scientific advice, (2) Fisheries Management, and (3) Capacity Development. A new state-of-the-art research vessel – R/V *Dr Fridtjof Nansen* – is an integral part of the programme and serves to conduct marine research with the participation of scientists from partner countries. It is the only marine research vessel to fly the United Nations flag.

CHALLENGE 4:
DEVELOP A SUSTAINABLE AND EQUITABLE OCEAN ECONOMY

- **SDG 14:** To achieve a ***sustainable*** blue economy, the health of the ocean must be central to the technology, innovation, and policy and management decisions associated with it. The marine transportation sector, fisheries and tourism industries have already proven to be critical components of the global economy but focusing on sustainable development and transformation of these industries will require engagement of diverse stakeholders across many sectors. Working together with the ocean science community, the Ocean Decade can contribute the knowledge and data to develop decision support tools for local communities, economists, private industries and governments.
- **Other key SDG interactions:**
  — **SDG 1 –** Support for a sustainable Blue Economy means the creation of new jobs and resources, particularly in SIDS and LDCs, which can contribute towards the Targets of 1.1 (eradicate extreme poverty) and 1.2 (reduce at least by half the proportion of men, women and children of all ages living in poverty in all its dimensions).
  — **SDG 12 –** A sustainable Blue Economy will require developing the implementation tools to monitor sustainable development impacts for sustainable tourism that creates jobs and promotes local culture and products (Target 12.b). The Ocean Decade also prioritises support for SIDS and LDCs to strengthen their scientific and technological capacity to move towards more sustainable patterns of consumption and production, specifically of marine resources (Target 12.a).
  — **SDG 7 –** Renewable energy from ocean-based technologies will be critical to filling the gaps to enhance the share of renewables in the global energy mix (Target 7.1), including development of offshore wind farms and tidal energy.

CHALLENGE 5:
UNLOCK OCEAN-BASED SOLUTIONS TO CLIMATE CHANGE

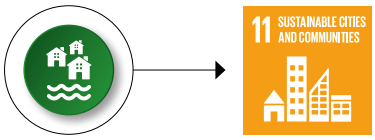
- **SDG 14:** The ocean and climate go hand in hand. The Ocean Decade will contribute data, knowledge and capacity to better understand and communicate the importance of the ocean-climate nexus. Filling these critical knowledge gaps with innovative science, marine technology, and observations – and focusing on ocean-based solutions to climate change – has great potential for positive impact on climate action and the 2030 Agenda. This knowledge will be relevant for numerous global policy frameworks, including the UNFCCC Paris Agreement. Addressing Challenge 5 is also transversal across the ten Ocean Decade Challenges.
- **Other key SDG interactions:**
  - **SDG 1 –** The ocean-climate nexus is inherently an issue of inequality, with developed countries contributing the most to greenhouse gas emissions and developing countries often threatened by the most severe impacts of climate change. By unlocking ocean-based solutions to climate change, the Ocean Decade will support actions that build the resilience of the poor and those in vulnerable situations and reduce their exposure and vulnerability to climate-related extreme events and other economic, social and environmental shocks and disasters (Target 1.5)
  - **SDG 13 –** A priority for the Ocean Decade is to elevate the role of the ocean in climate change adaptation and mitigation. Addressing Challenge 5 will provide the knowledge needed to inform global policy frameworks, specifically the UNFCCC Paris Agreement, on the integration of the ocean into climate change measures including national policies, strategies and planning (Target 13.2). Increasing ocean literacy within the context of climate change, through awareness raising, capacity development and early warning systems (Target 13.3), will also be critical to achieving SDG 13.

Case Study: A Decade Collaborative Centre focused on ocean-climate solutions

The Ocean Visions – U.N. Decade Collaborative Centre for Ocean-Climate Solutions

Launching in 2022, *the Ocean Visions – U.N. Decade Collaborative Centre for Ocean-Climate Solutions* is a partnership between Ocean Visions, Inc., Georgia Tech University, and the Georgia Aquarium, based in the United States, that will provide coordinated support on the development, testing, and scaling of ocean-based solutions such as mitigating the impacts of climate change on marine ecosystems and new technologies for carbon dioxide removal. This Collaborative Centre will not only contribute to our understanding of the ocean-climate nexus through coordination of endorsed Decade Programmes, but will also bring together diverse stakeholders to map out solutions through innovation, technology and unlocking new sources of funding.

CHALLENGE 6:
INCREASE COMMUNITY RESILIENCE TO OCEAN HAZARDS



- **SDG 14:** Enhanced coastal ocean observing systems, reliable and accessible data, and efforts to increase ocean literacy on ocean hazards has the potential to save lives and build resilient coastal communities. Coastal resilience can be achieved through the generation of new knowledge and tools for nature-based solutions that will increase the adaptative capacity of millions of vulnerable populations. Nature-based solutions will also positively reinforce the protection and restoration of important coastal and marine habitats, resulting in healthy ecosystems. With a safe ocean featured as one of the seven societal outcomes, the Western Tropical Atlantic Safe Ocean Working Group that emerged from the regional planning workshops, has proposed the development of a co-designed Integrated Multi Hazard Ocean Data and Forecast System (von Hillebrandt-Andrade et al., 2021).

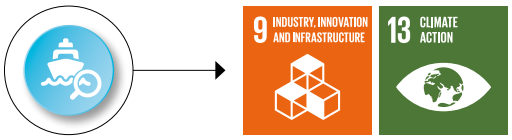
- **Other key SDG Interactions**
  - **SDG 11 –** By enhancing community resilience to ocean hazards, there is potential to significantly reduce the number of deaths and the number of people affected and substantially decrease the direct economic losses relative to global gross domestic product caused by natural and climate-related disasters (e.g., sea level rise), with a focus on protecting the poor and people in vulnerable situations (Target 11.5). Additionally, the work to support Challenge 6 can substantially increase the number of cities and human settlements adopting and implementing integrated policies and plans towards inclusion, resource efficiency, mitigation and adaptation to climate change, resilience to disasters, and develop and implement, in line with the Sendai Framework for Disaster Risk Reduction 2015-2030, holistic disaster risk management at all levels (Target 11.b).

Case Study: The Coastal Resilience Community of Practice

Communities of Practice within the Ocean Decade are designed to bring together proponents of endorsed Decade Actions working towards a common theme.

- Communities of Practice (CoP): Coastal Ecosystem and Community Resilience
- Lead Actions: The Global Estuaries Monitoring Programme, Ocean Cities, and Mega-Deltas
- The Coastal Ecosystem and Community Resilience CoP launched in 2021 and currently has 59 members from endorsed Decade Programmes, Projects, and Contributions. They focus on three core themes: 1) human interaction and connectivity; 2) coastal area fragmentation by the socio-economical environmental point of view as well as in terms of knowledge and best practice; and 3) engagement of society. Through collaboration in the digital space of the Global Stakeholder Forum and in person via the many diverse projects involved, they will positively contribute to increase understanding of the drivers and barriers to community resilience. Their role as a CoP also means that they will support the Decade Coordination Unit's efforts to engage relevant stakeholders in discussions of coastal resilience and encourage the development of Decade Actions to fill knowledge and solution gaps within their thematic focus.
- A Decade Collaborative Centre focused on Coastal Resilience in a Changing Climate, hosted by the University of Bologna, will launch in 2022 to support and facilitate this Community of Practice.

CHALLENGE 7:
EXPAND THE GLOBAL OCEAN OBSERVING SYSTEM



- **SDG 14:** Expanding and enhancing the global ocean observing system, and supporting regional observing networks, will provide reliable data that can inform all SDG 14 targets and indicators. One of the flagship Decade Programmes, Ocean Practices for the Decade (https://www.oceanbestpractices.org) led by the International Ocean Data and Information Exchange (IODE), is one example of initiatives that will be critical to supporting global participation in the Decade and addressing the SDG 14 indicators by establishing and coordinating the standardization of the ocean variables we observe.
- **Other key SDG interactions:**
  - **SDG 9 –** Expanding the global ocean observation system addresses target 9.1 which aims to develop infrastructure at regional scales supporting economic development and human well-being. Global ocean observations supported through regional scale initiatives through the Decade will help refine critical ocean infrastructure to ensure the safety of communities around the world.
  - **SDG 13 –** Ocean observations underpin the ability of communities to adapt to ocean related disasters effectively by providing accurate and up to date knowledge of the ocean climate nexus. This helps achieve SDG target 13.1 which aims to enhance community capacity to respond to climate disasters.

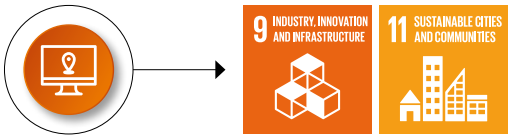
Case Study of UN-led Decade Actions

Programme: Ocean Observing Co-Design: evolving ocean observing for a sustainable future

Ocean Observing Co-design is lead by the Global Ocean Observing System (GOOS) programme within the IOC-UNESCO. This initiative will support the development of fit-for-purpose observing systems that involves working with stakeholders across diverse sectors of society to improve the benefits 1) Better track the current state and future variability of the ocean 2) Predict and warn more skillfully 3) Manage ocean resources 4) Empower society to adapt to change and 5) Assess the impact of action towards a sustainable ocean.

https://www.goosocean.org

CHALLENGE 8:
CREATE A DIGITAL REPRESENTATION OF THE OCEAN



- **SDG 14 –** Using the latest advances in technology, including ocean models, artificial intelligence, machine learning and high-performance computing, will ensure that we not only better understand the ocean, but improve climate projections, make informed decisions that influence the health of marine ecosystems, enhance the sustainable blue economy, and contribute to developing climate mitigation and adaptation strategies for coastal communities.
- **Other key SDG interactions:**
  - **SDG 9 –** Industry engagement with the Ocean Decade is rapidly gaining momentum in the development of data tools. This will support the achievement of SDG target 9.2 which aims to raise industry's share of employment, particularly in least developed countries, another priority area for the Decade.
  - **SDG 11 –** Open access to near-real time representations of the ocean and coastal zone will provide the reliable data needed by communities to make informed decisions that are necessary to minimize loss and damage.

Case Study of Decade Actions: Creating Digital Twins

Programme: Digital Twins of the Ocean (DITTO)
Decade Challenges: 1, 2, 3, 4, 5 ,6 ,7, 8, 9, 10
Top 3 SDGs: 9, 14, 17

DITTO will support the development of digital twins of the ocean as "a realistic digital representations of assets, processes or systems in the built or natural environment". This will be accomplished by establishing and advancing a digital framework on which all marine data, satellite data and reanalyzes, modelling and simulation along with high-performance computing (HPC) capacities, artificial intelligence (AI) algorithms and specialized tools including relevant best practices will form a new globally shared capacity to access, manipulate, analyze, visualize and effectively use marine information. DITTO will enable users and partners to create their own digital twin of the ocean to answer "what if" questions in support of a wide range of use in the context of ocean development addressing issues such as development of a sustainable Blue Economy and effective Maritime Spatial Planning.

CHALLENGE 9:
SKILLS, KNOWLEDGE, AND TECHNOLOGY FOR ALL (CAPACITY DEVELOPMENT)

- **SDG 14:** The 2020 Global Ocean Science Report revealed the lack of resources dedicated to ocean science and highlighted the unequal distribution of capacity (IOC-UNESCO, 2020a). Supporting capacity development is cross-cutting and essential to the achievement of SDG 14. The Decade Action Framework was designed with this in mind, aiming to increase scientific knowledge, develop research capacity and transfer marine technology, in order to improve ocean health and to enhance the contribution of marine biodiversity to the development of developing countries, in particular SIDS and LDCs. The Ocean Decade will work with national organizations to mobilise resources to increase the proportion of total research budgets allocated to research in the field of marine technology (indicator 14.a.1). The IOC-UNESCO is the custodian agency for indicator 14.a.1 and will capitalize on the Ocean Decade framework and emerging interests to facilitate new partnerships, support capacity building initiatives, and support the data collection to report progress on this indicator.
- **Other key SDG interactions:**
  - **SDG 12 –** Addresses Target 12.8 which aims to ensure that people everywhere have the relevant information and awareness for sustainable development and lifestyles in harmony with nature
  - **SDG 17 –** Building capacity within the ocean science community around the world, through coordinated efforts to monitor and report on SDG 14 targets and indicators, enhanced ocean

literacy, and marine technology transfer, will ensure global participation is possible for the implementation of the Ocean Decade and achieving sustainable development of the ocean and society.

Case Study of Decade Actions: Building Capacity for the Ocean Decade

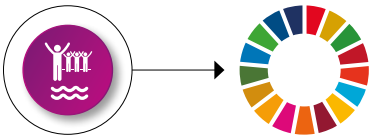
Programme: Ocean Voices

Decade Challenges: 4, 10

Top 3 SDGs: 3, 10, 14

The contribution of ocean science to sustainable development is determined by people. Understanding the actors involved, their knowledge systems, culture and well-being, and how power dynamics and decision-making processes influence our oceans is crucial to achieve the goals of the Decade. There is a need to ensure knowledge, strategy and governance frameworks to enable all to participate in, contribute to and benefit equitably from the Decade. The Oceans Voices program will conduct research, incubate ideas, facilitate critical discussions and convene capacity building partnerships to identify barriers, pathways and enabling conditions for equity in the Decade.

CHALLENGE 10:
CHANGE HUMANITY'S RELATIONSHIP WITH THE OCEAN



- Challenge 10 has the potential for positive impact across all SDGs. By changing the way we think about and value the ocean, we can harness the transversal nature of the ocean for the benefit of society while protecting and restoring marine ecosystems for future generations.
- **SDG Interactions:** Challenge 10 will result in improved ocean literacy, new technologies, science-informed policies and management decisions. It will also support new opportunities in the ocean sciences that will result in better diversity, equity and inclusion.

Case Study of Decade Actions: SDG interaction highlight SDG 14-SDG 5

Programme: Empowering Women for the United Nations Decade of Ocean Science for Sustainable Development

Decade Challenges: 2, 4, 9, 10

Empowering Women for the Ocean Decade aims to advance an alternative, inclusive model to enable more representative ocean science knowledge production to ensure gender equality and the empowerment of women of different ages and backgrounds at all levels. The Programme's research will identify outcomes which can be used in institutional capacity building and awareness raising. This includes the development of a baseline study on women's empowerment in selected intergovernmental organizations and non-governmental organizations concerned with ocean affairs. Research findings will identify key barriers and good practices which can inform the proposed Strategy and Action Plan. This will contribute to delivering equal opportunities for full participation and leadership by women at all levels of ocean science under the Ocean Decade. The Programme will enhance ongoing collaborations with partners to deliver substantive research outputs contributing to transformative actions.

The flagship Decade Actions will set the stage for action over the course of the decade. Proponents of Decade Actions (programmes and projects) were asked to identify which of the Decade Challenges their endorsed Action aimed to address. An initial assessment of this data reveals that there are several Decade Challenges, and therefore the potential to address numerous SDGs, that will require additional focus to address. Regular scoping exercises will be carried out within the Ocean Decade community to identify where gaps remain in terms of contribution to the SDGs and this information will be used to develop the scope of future Calls for Decade Actions, track progress against the Challenges and allow for ongoing refinement and evolution of the Ocean Decade Action Framework over time.

An Enabling Environment for Achieving Ocean Science Aspects of SDG 14 and the 2030 Agenda

To overcome the barriers of achieving SDG 14, the Ocean Decade was designed to create an enabling environment that will raise awareness, mobilise resources and inspire the transformative ocean science that will be co-designed with all stakeholders, including policy and decisions-makers, which will be vital in fostering the political will needed for action. The Ocean Decade will also provide the knowledge to fill critical gaps in achieving numerous global policy frameworks, including the Paris Agreement. The Ocean Decade will also support initiatives that break down systemic barriers that will allow weaving together of different knowledge.

Finance for ocean science: The Ocean Decade has developed several mechanisms to support resource mobilization efforts that connect Member States, philanthropic foundations, and the private sector with endorsed Decade Actions that address the ten Decade Challenges, including the Ocean Decade Alliance and the Foundations Dialogue. Further, the Ocean Decade has established a mechanism through which partners can co-sponsor Calls for Decade Actions to support resource mobilisation around specific Decade Challenges. There are signs of success emerging as a result of the collective efforts of partners and key stakeholders. At the close of the first year of the Ocean Decade (2021), $840 million was secured in funding by endorsed Decade Actions and over $15 million was leveraged in funding to support new Decade Actions.

- Key Mechanisms for Resource Mobilisation:
  - **Ocean Decade Alliance**: The Ocean Decade Alliance is a network of preeminent members of society that use their influence and resources to support the Ocean Decade. The Alliance also includes Member States, foundations, institutional members (UN entities), and the private sector.
  - **The Foundations Dialogue:** Launched in February 2020 in Copenhagen, the Netherlands, the Foundations Dialogue brings together more than 25 foundations from Africa, Europe, Oceania, North America, and South America to discuss how to effectively mobilise resources for Decade priorities. Since the Copenhagen meeting, the Foundations Dialogue Members have continued to work together foster collaborations amongst diverse partners that have the potential to generate and disseminate the transformational knowledge and scientific capacity needed to address Decade Challenges.
  - **Co-Sponsored Call for Decade Actions:** co-sponsored Calls for Decade Actions are initiatives that align with the Ocean Decade objectives and challenges and can provide both funding and endorsement of Decade Actions. These included:
    1. The AXA Research Fund committed €1 million to launch a call for proposals to fund early career researchers, co-developed with the Decade Coordination Unit.
    2. The Meer Wissen Initiative African-German Partnership for Marine Research sponsored call for proposals focused on the co-design phase of projects for "Nature-based solutions in Africa."
    3. ArcticNet, in partnership with IOC-UNESCO and Mitacs, issued a call for proposals that was partially funded by Fisheries and Oceans Canada (DFO) to initiate new projects that are relevant to ArcticNet's mandate and address emerging science priorities that support a self-determined and sustainable Canadian North.

Political and societal engagement: Policies at the National level are critical to implementation of sustainability measures. The role of Member States in the Ocean Decade will be important in funding many of these initiatives and can support efforts to translate the science generated through the Decade into national policies. Member States can contribute to the Decade through both in-kind and financial support of endorsed Actions. A growing number of countries are joining the Ocean Decade Alliance. The Ocean Decade is also facilitating coordination at the national level through the establishment of National Decade Committees to support endorsed Decade Actions, catalyse new actions, and to generate knowledge that can potentially inform policies at the national level, which ultimately influence global policy frameworks (Figure 5). The Ocean Decade is prioritizing the expansion of these National Committees, with a specific emphasis on supporting their formation in SIDS and LDCs.

Focusing efforts on ocean literacy campaigns will improve engagement and uptake of ocean science by society. The IOC-UNESCO has been a leader in Ocean Literacy efforts, with great accomplishments occurring under the Ocean Literacy for All campaign (www.oceanliteracy.unesco.org). These efforts are being translated in the Ocean Decade framework under the Ocean Literacy With All (OLWA), one of the UN-led endorsed Decade Programmes. The OLWA programme focuses on capacity development and will support projects that reach diverse networks of stakeholders, identify and address ocean literacy knowledge gaps, and support ocean literacy efforts throughout all of the Ocean Decade programmes.

Figure 5. Established National Decade Committees in 2021–2022.

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Overcoming systemic barriers to transdisciplinary and participatory research: Placing emphasis on the co-design of ocean science will require a change in the way we think about and fund science. As previously noted, new funding models are being introduced through partnerships with philanthropic, national and regional organizations that will provide support for this critical phase of transdisciplinary research. The Ocean Decade will also provide training opportunities to the research community on how to effectively design and implement a co-designed ocean science project. Other endorsed programmes are leading their own workshops and developing training materials tailored thematically to support this shift in research models.

The Ocean Decade also recognizes that indigenous and local knowledge has been used for centuries to millennia to sustainably manage marine and coastal ecosystems, providing the resources for these communities. By recognizing these knowledge systems as integral to ocean science, and as indigenous science in its own right, there could be greater engagement and achievement of other SDGs. The UN has recognized that the inclusion of indigenous knowledge systems in sustainable development also has transversal characteristics to achieve all SDGs (United Nations, 2022). The Ocean Decade promotes opportunities to lead the way in truly transformative approaches, examples include a) approaches to designing marine protected areas, b) supporting food security and traditional fishing practices, and c) working with communities to co-design and mobilise resources to support indigenous observing networks for monitoring and reporting on the effects of climate change including coastal erosion, human health and well-being, and shifting species ranges. An Indigenous and Local Knowledge Framework for the Ocean Decade was proposed in the first year to outline guiding principles for ethical engagement, serve as a knowledge hub for the ocean science community, and to elevate the voices of indigenous peoples and local communities on the global stage.

Case Study: Weaving traditional knowledge with integrated ocean management in Pacific SIDS

Programme: Pacific solutions to save our ocean

Challenges: 2, 3, 4, 5, 9, 10

Top 3 SDGs: 13, 14, 17

Pacific Solutions to Save Our Ocean, led by the regional organization Pacific Community (SPC), aims to leverage the spate of national ocean policies launched in the Pacific in recent years to address the need for greater integrated ocean management. This Decade Programme seeks to increase scientific capacity and create opportunities for ocean science to feed into decision making. This will be achieved by focusing on three major components: 1) Develop robust legal and governance frameworks to implement national ocean policy, 2) Improve decision support systems so that policies are informed by science, 3) Integrate Pacific traditional knowledge, culture, and wisdom to support inclusive, consultative decision making and improve decision support systems.

As custodians of the world's largest marine ecosystems, the Blue Pacific Continent, SPC and its island members seek to save the ocean and provide solutions to some of our existential challenges including climate change.

Conclusions

Now is the time to act. As the last decade of implementation of the 2030 Agenda overlaps with the Ocean Decade, there is a unique opportunity to create synergies and fill critical knowledge gaps that will hinder the progress on the Goals. The UN Ocean Conference, and the annual ocean-climate dialogue set out in the Glasgow Climate Pact, must consider the most recent developments in the science, working with and directly engaging the ocean science community, inclusive of Indigenous and local knowledge holders. The ocean needs to be central to the discussions on climate, the science needs to inform decisions related to biodiversity on the high seas, and the science must be co-designed to establish marine protected areas that are accessible and have economic benefit.

Looking ahead. To continue promoting the transformative ocean science that is already emerging from the Ocean Decade, and to highlight the strategic ambitions set out to inform global policy frameworks, the presence of the Ocean Decade at important gatherings will be instrumental. Opportunities such as those as the upcoming UNFCCC COP 27 meeting (Egypt, 2022) will provide an opportunity to promote ocean-based solutions to climate change and inclusion of the ocean into national climate policies, bringing together scientists and policy makers within their respective countries to coalesce around solutions. At the upcoming Convention on Biodiversity (CBD) COP 15, the Ocean Decade will highlight the science and resource needs that will inform the indicators associated with the post-2020 global biodiversity framework.

Measuring progress. The Ocean Decade is in the process of establishing a formal monitoring and evaluation process that will assess outcomes and performance, including how it has contributed to SDG 14 and other SDGs. By aligning these metrics with the SDG 14 targets, and expanding beyond these targets, the Ocean Decade will showcase the benefits of investing in transformative ocean science for society, the economy, and the environment. This monitoring and evaluation process, and progress towards the achieving the Goals, will be used to inform future Calls for Decade Actions to ensure the sustainable development of the ocean for the benefit of all of society and for the ocean.

Bibliography

Arctic Council. 2021. Information Brief: *Indigenous Food Security in the Arctic Implications of a Changing Ocean.* Protection of the Marine Environment (ed.).

Bank, M. S., Duarte, C. M. & Sonne, C. 2022. Intergovernmental Panel on Blue Foods in Support of Sustainable Development and Nutritional Security. *Environmental Science & Technology*, 56, 5302-5305.

Claudet, J., Bopp, L., Cheung, W. W. L., Devillers, R., Escobar-Briones, E., Haugan, P., Heymans, J. J., Masson-Delmotte, V., Matz-Lück, N., Miloslavich, P., Mullineaux, L., Visbeck, M., Watson, R., Zivian, A. M., Ansorge, I., Araujo, M., Aricò, S., Bailly, D., Barbière, J., Barnerias, C., Bowler, C., Brun, V., Cazenave, A., Diver, C., Euzen, A., Gaye, A. T., Hilmi, N., Ménard, F., Moulin, C., Muñoz, N. P., Parmentier, R., Pebayle, A., Pörtner, H.-O., Osvaldina, S., Ricard, P., Santos, R. S., Sicre, M.-A., Thiébault, S., Thiele, T., Troublé, R., Turra, A., Uku, J. & Gaill, F. 2020. A Roadmap for Using the UN Decade of Ocean Science for Sustainable Development in Support of Science, Policy, and Action. *One Earth*, 2, 34-42.

Di Franco, A., Thiriet, P., Di Carlo, G., Dimitriadis, C., Francour, P., Gutiérrez, N. L., Jeudy de Grissac, A., Koutsoubas, D., Milazzo, M., Otero, M. d. M., Piante, C., Plass-Johnson, J., Sainz-Trapaga, S., Santarossa, L., Tudela, S. & Guidetti, P. 2016. Five key attributes can increase marine protected areas performance for small-scale fisheries management. *Scientific Reports*, 6, 38135.

FAO. 2013. *Good practice policies to eliminate gender inequalities in fish value chains*, Food and Agriculture Organization of the United Nations.

FAO. 2021. The White/Wiphala Paper on Indigenous Peoples' food systems. Rome.

Hoegh-Guldberg, O., Northrop, E. & Lubchenco, J. 2019. The ocean is key to achieving climate and societal goals. *Science*, 365, 1372-1374.

IOC-UNESCO. 2020a. *Global Ocean Science Report 2020 – Charting Capacity for Ocean Sustainability*. K. Insensee (ed.), Paris, UNESCO.

IOC-UNESCO. 2020b. T*he Transformative Role of Foundations in the Ocean Decade [UN Decade of Ocean Science for Sustainable Development (2021–2030)]*. UNESCO (eds.), Paris, (The Ocean Decade Series, 16; IOC Brochure 2020-12).

IOC-UNESCO. 2022. *Multiple Ocean Stressors: A Scientific Summary for Policy Makers*. P.W. Boyd et al. (eds). Paris, UNESCO. 20 pp (IOC Information Series, 1404) doi:10.25607/OBP-1724

IPCC. 2019. Summary for Policymakers. In: *IPCC Special Report on the Ocean and Cryosphere in a Changing Climate*. H.-O. Pörtner, D. C. R., V. Masson-Delmotte, P. Zhai, M. Tignor, E. Poloczanska, K. Mintenbeck, A. Alegri´a, M. Nicolai, A. Okem, J. Petzold, B. Rama, N.M. Weyer (ed.).

IPCC. 2022. Summary for Policymakers. In: *Climate Change 2022: Mitigation of Climate Change. Contribution of Working Group III to the Sixth Assessment Report of the Intergovernmental Panel on Climate Change*. P.R. Shukla, J. Skea, R. Slade, A. Al Khourdajie, R. van Diemen, D. McCollum, M. Pathak, S. Some, P. Vyas, R. Fradera, M. Belkacemi, A. Hasija, G. Lisboa, S. Luz, J. Malley (eds.). Cambridge University Press, Cambridge, UK and New York, NY, USA.

Jaspars, M., De Pascale, D., Andersen, J. H., Reyes, F., Crawford, A. D. & Ianora, A. 2016. The marine biodiscovery pipeline and ocean medicines of tomorrow. *Journal of the Marine Biological Association of the United Kingdom*, 96, 151-158.

Karr, K. A., Fujita, R., Carcamo, R., Epstein, L., Foley, J. R., Fraire-Cervantes, J. A., Gongora, M., Gonzalez-Cuellar, O. T., Granados-Dieseldorff, P., Guirjen, J., Weaver, A. H., Licón-González, H., Litsinger, E., Maaz, J., Mancao, R., Miller, V., Ortiz-Rodriguez, R., Plomozo-Lugo, T., Rodriguez-Harker, L. F., Rodríguez-Van Dyck, S., Stavrinaky, A., Villanueva-Aznar, C., Wade, B., Whittle, D. & Kritzer, J. P. 2017. Integrating Science-Based Co-management, Partnerships, Participatory Processes and Stewardship Incentives to Improve the Performance of Small-Scale Fisheries. *Frontiers in Marine Science*, 4.

Lawrence, M. G., Williams, S., Nanz, P. & Renn, O. 2022. Characteristics, potentials, and challenges of transdisciplinary research. *One Earth*, 5, 44-61.

Mercator Ocean International. 2022. *Digital Twin Ocean* [Online]. Available: http://www.fao.org/3/i3553e/i3553e.pdf [Accessed 08 April 2022].

National Academies of Sciences, Engineering, and Medicine. 2022. *A Research Strategy for Ocean-based Carbon Dioxide Removal and Sequestration*. Washington, DC: The National Academies Press.

OECD. 2022. *The SDG Financing Lab* [Online]. Available: https://sdg-financing-lab.oecd.org/ [Accessed 04 April 2022].

Ortuño Crespo, G., Mossop, J., Dunn, D., Gjerde, K., Hazen, E., Reygondeau, G., Warner, R., Tittensor, D. & Halpin, P. 2020. Beyond static spatial management: Scientific and legal considerations for dynamic management in the high seas. *Marine Policy*, 122, 104102.

Pepi, M. & Focardi, S. 2021. Antibiotic-Resistant Bacteria in Aquaculture and Climate Change: A Challenge for Health in the Mediterranean Area. *International journal of environmental research and public health*, 18, 5723.

Rose, J. M., Gast, R. J., Bogomolni, A., Ellis, J. C., Lentell, B. J., Touhey, K. & Moore, M. 2009. Occurrence and patterns of antibiotic resistance in vertebrates off the Northeastern United States coast. *FEMS microbiology ecology*, 67, 421-431.

Schmidt, S., Neumann, B., Waweru, Y., Durussel, C., Unger, S. & Visbeck, M. 2017. SDG 14-conserve and sustainable

use the oceans, seas and marine resources for sustainable development. *A guide to SDG interactions: from science to implementation*. International Council for Science (ICSU).

Scotland, P. 2022. *Why we need to tackle the ocean funding crisis* [Online]. Available: https://ocean.economist.com/blue-finance/articles/why-we-need-to-tackle-the-ocean-funding-crisis [Accessed 04 April 2022].

Singh, G. G., Cisneros-Montemayor, A. M., Swartz, W., Cheung, W., Guy, J. A., Kenny, T.-A., McOwen, C. J., Asch, R., Geffert, J. L., Wabnitz, C. C. C., Sumaila, R., Hanich, Q. & Ota, Y. 2018. A rapid assessment of co-benefits and trade-offs among Sustainable Development Goals. *Marine Policy*, 93, 223-231.

Smith, M., Love, D. C., Rochman, C. M. & Neff, R. A. 2018. Microplastics in Seafood and the Implications for Human Health. *Current environmental health reports*, 5, 375-386.

Sturesson, A., Weitz, N. & Persson, Ã. 2018. SDG 14: Life Below Water. *A Review of Research Needs. Technical annex to the Formas report Research for Agenda 2030*.

UNCTAD. 2016. *Oceans Economy and Ecosystem services: sustainable fisheries and coastal tourism. Side event to the CBD COP 13* [Online]. Available: https://unctad.org/es/node/25793 [Accessed 31 March 2022].

UNESCO-IOC. 2021. The United Nations Decade of Ocean Science for Sustainable Development (2021–2030) *Implementation Plan*. UNESCO (ed.), Paris (IOC Ocean Decade Series, 20.)

UNFCCC. 2020. Policy Brief: *Technologies for Averting, Minimizing and Addressing Loss and Damage in Coastal Zones*. Executive Committee of the Warsaw International Mechanism for Loss and Damage, Technology Executive Committee (ed.).

United Nations. 2017. 2017 *HLPF thematic review of SDG 14: conserve and sustainably use the oceans, seas and marine resources for sustainable development* [Online]. New York: High-level Political Forum. Available: https://sustainabledevelopment.un.org/content/documents/14375SDG14format-revOD.pdf [Accessed 04 May 2022].

United Nations. 2021. *Sustainable Blue Foods are Vital to Global Food Security* [Online]. Available: https://sdgs.un.org/news/sustainable-blue-foods-are-vital-global-food-security-33148 [Accessed 04 May 2022].

United Nations. 2022. *Indigenous Peoples and the 2030 Agenda* [Online]. Available: https://www.un.org/development/desa/indigenouspeoples/focus-areas/post-2015-agenda/the-sustainable-development-goals-sdgs-and-indigenous.html, [Accessed 04 April 2022].

von Hillebrandt-Andrade, C., Blythe-Mallett, A. & Escobar-Briones, E. 2021. Co-Designing a safe ocean in the Western Tropical Atlantic within the framework of the UN Decade of Ocean Science for Sustainable Development. *Ocean and Coastal Research*, 69.

Winther, J. G., Dai, M., Rist, T., Hoel, A. H., Li, Y., Trice, A., Morrissey, K., Juinio-Meñez, M. A., Fernandes, L., Unger, S., Scarano, F. R., Halpin, P. & Whitehouse, S. 2020. Integrated ocean management for a sustainable ocean economy. *Nature ecology & evolution*, 4, 1451-1458.

World Health Organization. 2019. *Health, the global ocean and marine resources:* policy brief. Copenhagen: World Health Organization. Regional Office for Europe.

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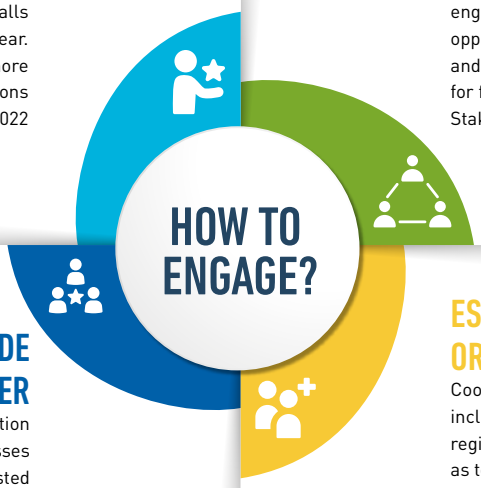

LEAD OR PARTICIPATE IN A DECADE ACTION

Decade Actions will be implemented by a wide range of proponents throughout the Decade. Regular Calls for Actions will be released twice per year. Visit www.oceandecade.org to learn more about the Call for Decade Actions No 03/2022

JOIN THE GLOBAL STAKEHOLDER FORUM

The Global Stakeholder Forum will serve as the primary engagement mechanism for the Ocean Decade, announcing opportunities for funding and partnerships, workshops and trainings, meetings, conferences, and input on calls for future Decade Actions. Sign up today to join the Global Stakeholder Forum at forum.oceandecade.org

JOIN THE OCEAN DECADE EXPERT ROSTER

The Expert Roster will assist the Decade Coordination Unit with strategic, technical and review processes throughout the life of the Ocean Decade. Interested individuals can find more information on www.oceandecade.org/expert-roster

ESTABLISH OR JOIN A REGIONAL OR NATIONAL DECADE COMMITTEE

Coordination at the national and regional level will foster inclusion, engage national stakeholders and facilitate regional and national contributions to the Decade, as well as to promote awareness and interest. These voluntary and multi-stakeholder platforms will be key in linking national action to the international Ocean Decade framework. Learn more about national and regional groups on www.oceandecade.org

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